

BOARD GAMING AI AGENT

ABSTRACT

A Board Gaming AI Agent is an intelligent system designed to simulate human-like decision-making in strategic board games. This project focuses on developing an AI-powered agent capable of playing board games efficiently by analyzing game states, predicting opponent moves, and selecting optimal actions. The agent utilizes techniques such as heuristic search, game tree exploration (e.g., Minimax algorithm with Alpha-Beta pruning), and machine learning approaches like reinforcement learning to improve its performance over time.

The system is designed to support multiple board games, making it adaptable and scalable across different rule sets and complexities. By incorporating state evaluation functions and adaptive learning mechanisms, the AI agent can make strategic decisions under uncertainty and dynamic conditions. Additionally, the project aims to provide a user-friendly interface where players can interact with the AI, enabling both competitive gameplay and learning opportunities.